Structures

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Quiz 1: Structure

```
struct { int a; double b; } x;
```

a) x is a variable having a and b as components

b) x is a type having a and b as components

c) x->a is an int variable

d) None of these

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Quiz 2: Structure

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typedef struct { int a; double b; } x;
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Quiz 3: Structure

```
struct what { char ch; int n; } ;
```

if sizeof int is 4 bytes and char is 1 byte,

sizeof this structure is

a) 5

b) 8

c) Depends on the compiler

d) None of these

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Quiz 4: Structure

```
struct what { int a[10]; double b; } c[5];
```

How to refer to 5th element of a in the 2nd element of c counting from 0?

a) c[1].a[4]

b) c[1]->a[4]

c) c.a[1][4]

d) C[1][4].a

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Quiz 5: Structure

```
struct what { int a; double b; } a = {1279, 3.14}, b;

b = a;
```

a) Syntax error; structures cannot be assigned

b) The variable b will point to a

c) The variable b gets a fieldwise copy of b

d) None of these

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Quiz 5: Structure

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struct what { int a; double b; } a = {1279, 3.14}, b;

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c) The variable b gets a fieldwise copy of b

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Structure

- A structure has one or more components
- components are generally heterogeneous
- components have names - structure has named components
    - in case of array, the components are computed based on the value of an expression
- offset to each component is fixed at compile time
- supports random access. Time required to access the component does not depend on its position in the layout of the structure.

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Structure

- The size of a structure depends on implementation.
- The size of structure is at least equal to the sum of the sizes of data members.
- A structure variable stands for the whole structure.
- It does not degenerate to a pointer as is the case with arrays.
- Structures are assignment compatible.
- Conceptually, the compiler generates the code for memberwise copy.
- Structures can be assigned even if the structure contains an array.

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THANK YOU

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